Civil Engineering Data Analytics and AI

Introduction

Instructor

Heeseung Shon, Ph.D.

Course Objectives

❑ **This course aims to develop students' knowledge and skills in:**

1 Mathematical foundations and intuition for AI

- Understand key mathematical concepts behind data-driven models, including vectors and matrices, linear transformations, derivatives, and regularization.
- Explain how these concepts support dimensionality reduction, model training, and the prevention of overfitting.
- Use these ideas to interpret model behavior and results in civil engineering applications.

2 Modeling methods and model development

- Understand a range of modeling methods, including linear regression, decision trees and ensembles, support vector machines, k-means clustering, and neural networks.
- Select methods that are appropriate for the data, engineering problem, and analytical objective.
- Develop and compare models while addressing overfitting, bias, and uncertainty through regularization and cross-validation.

3 Data mining and reproducible practice with Python

- Preprocess real-world data by imputing missing values, handling outliers, and transforming variables.
- Evaluate models using appropriate metrics such as root-mean-square error, precision, and recall.
- Implement, document, and communicate a reproducible data-analysis workflow in Python.

Goal: Use data and AI responsibly to solve civil engineering problems and create value for society.

Textbook and References

- Lecture Note
- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow (Aurelien Geron)
- Deep Learning (Goodfellow, Bengio, Courville)

Class Contents

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Data Mining for Civil Engineering AI

Applied Math and Machine Learning Basics

Regression Models

Support Vector Machines

Decision Trees and Ensemble Learning

Clustering

Dimensionality Reduction

Neural Networks and Deep Learning

CNNs, RNNs, and GNNs

Model Evaluation and Generalization

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Introduction to AI in Civil Engineering

Smart Cities and Digital Twins

❑ Smart Cities

➢ A smart city uses **IoT sensors** and technology to **connect** components across a city to derive **data** and improve the lives of citizens and visitors.

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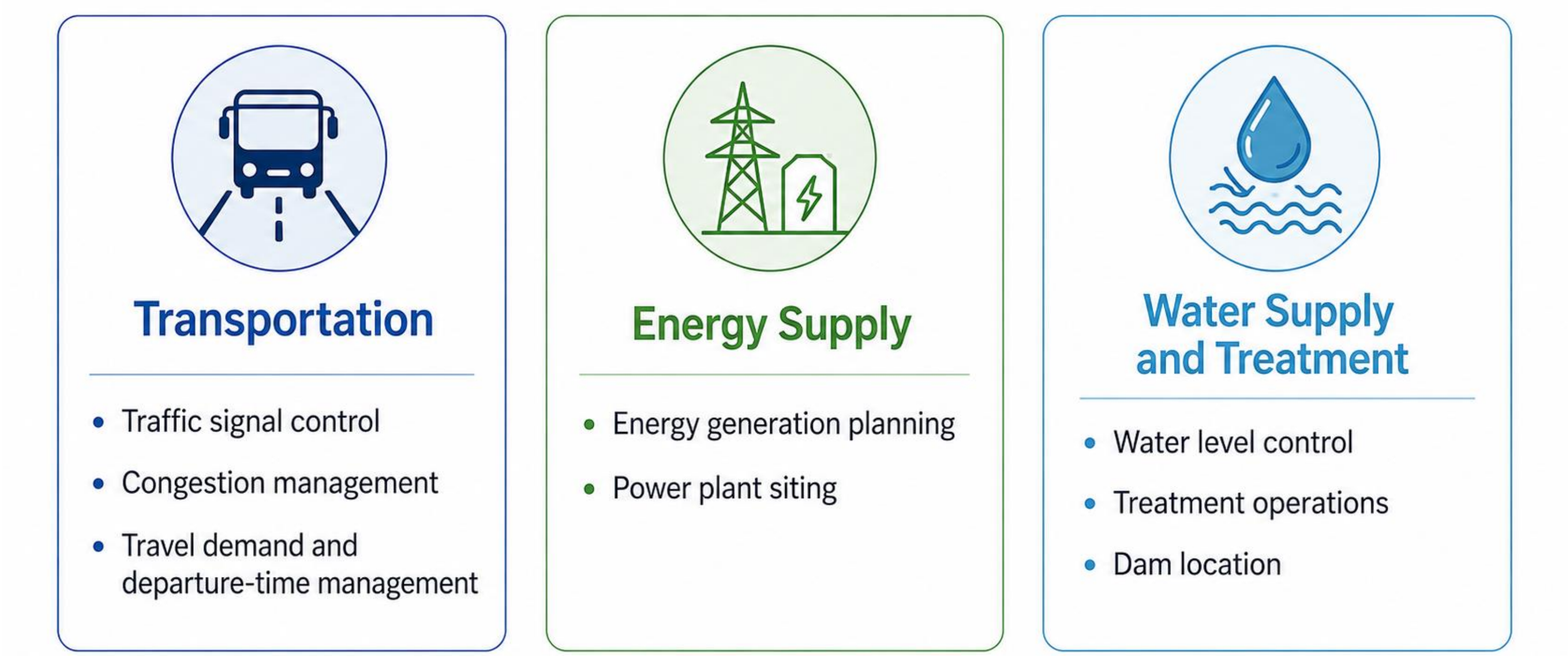
Smart Cities and Digital Twins

❑ Why Smart Cities Matter to Civil Engineers

- ➢ A smart city is a system of interconnected infrastructures.
- ➢ Civil engineers make important decisions across these systems.

❑ Examples



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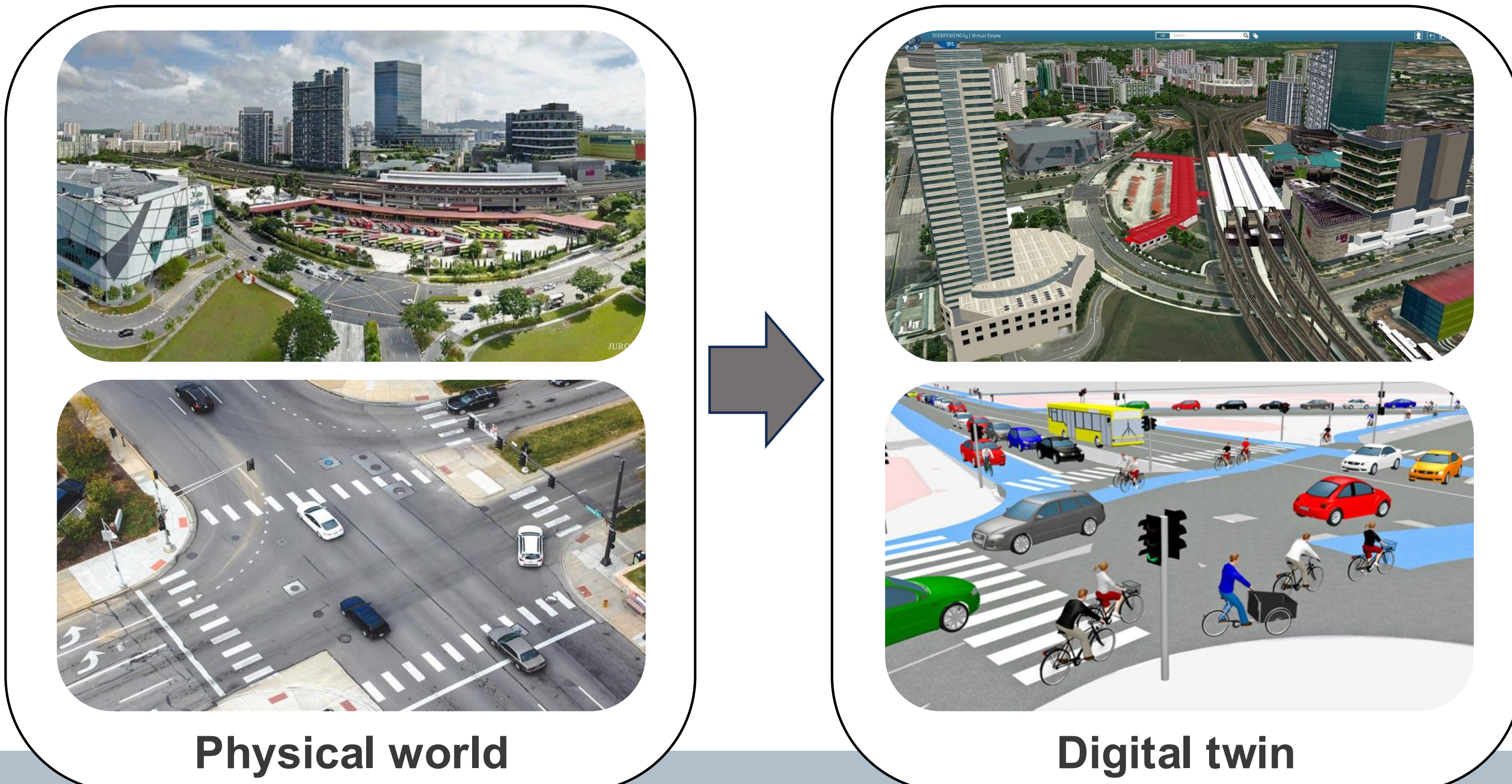
Smart Cities and Digital Twins

❑ Digital Twins in Civil Engineering

➢ A digital twin is **a dynamic virtual representation** of a physical system, continuously updated with real-world data.



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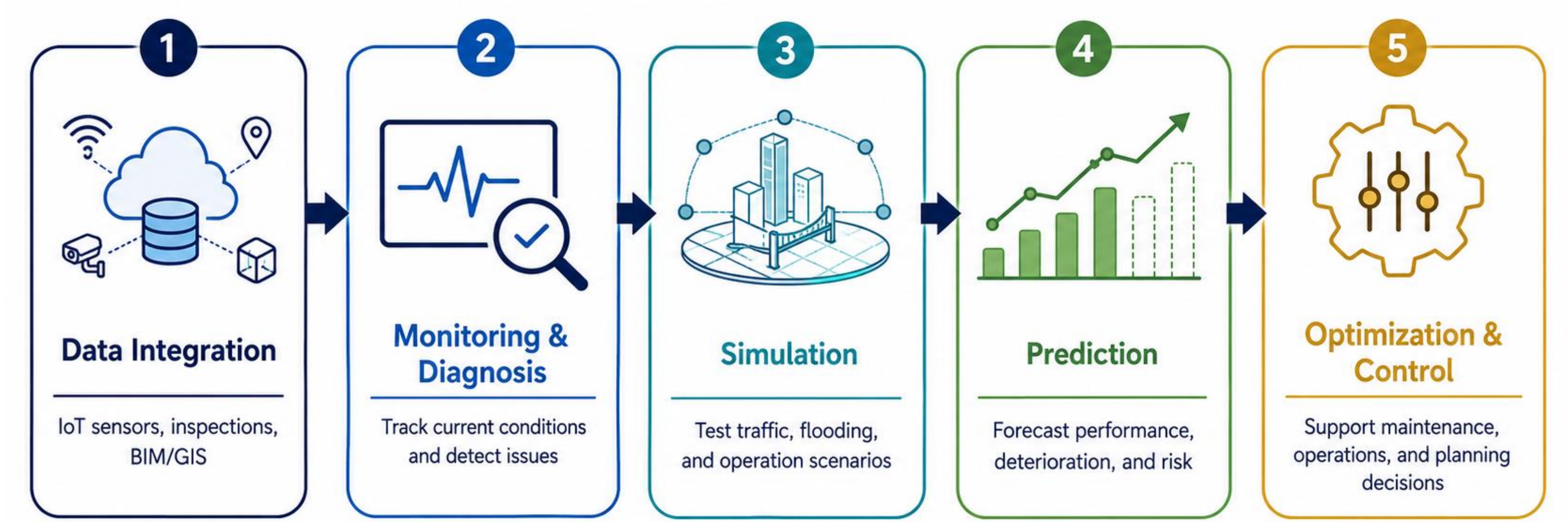
Smart Cities and Digital Twins

❑ Digital Twins in Civil Engineering

➢ Digital twins are used throughout the **infrastructure lifecycle** to **simulate performance**, **predict future conditions**, and **optimize engineering decisions**.



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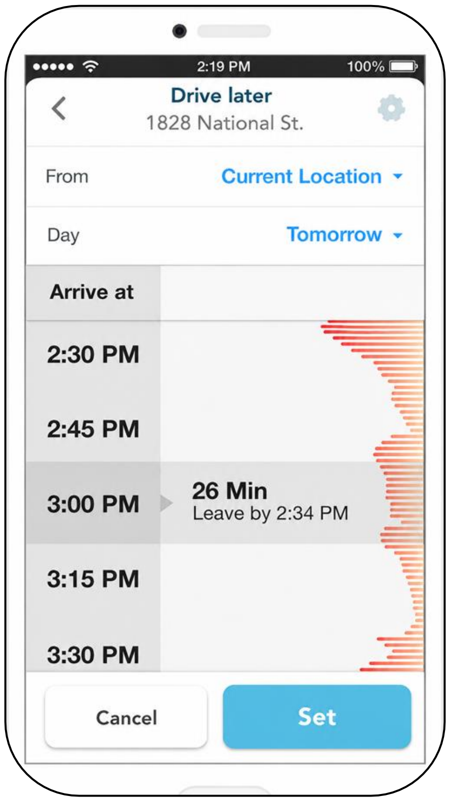
AI in Transportation Engineering

❑ Example: Travel Time Prediction

Driving travel-time prediction

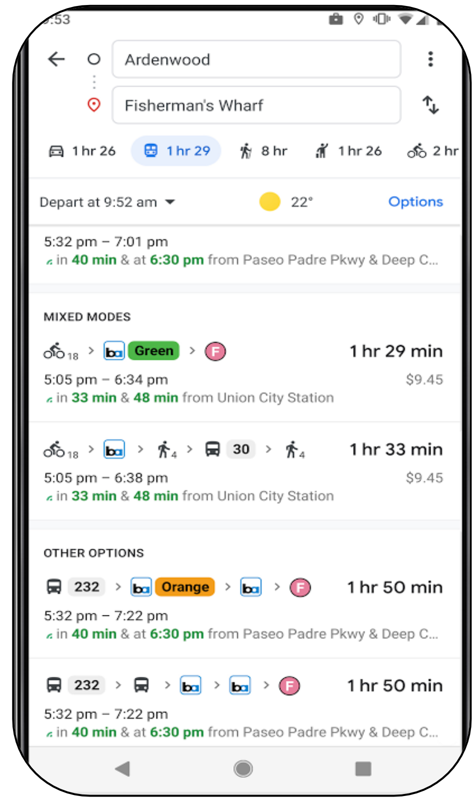


Ex) Waze, Google Maps

Transit arrival-time prediction

Ex) NJ Transit

Multimodal travel-time prediction



Ex) Google Maps

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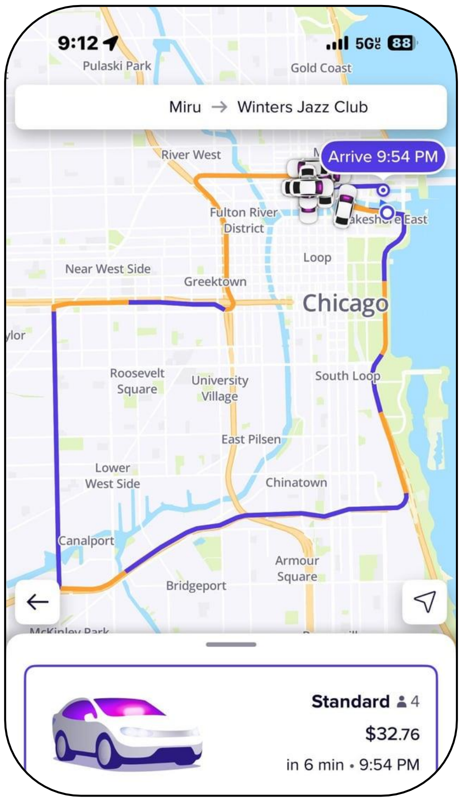
AI in Transportation Engineering

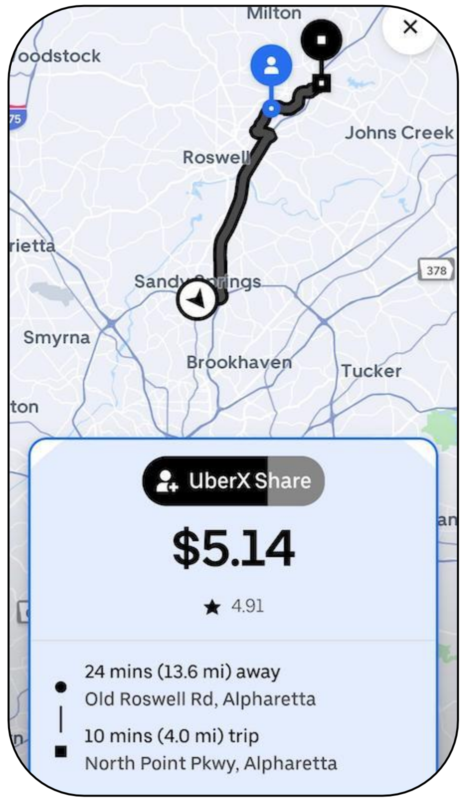
❑ Example: AI-Enhanced Route Optimization

Individual route optimization



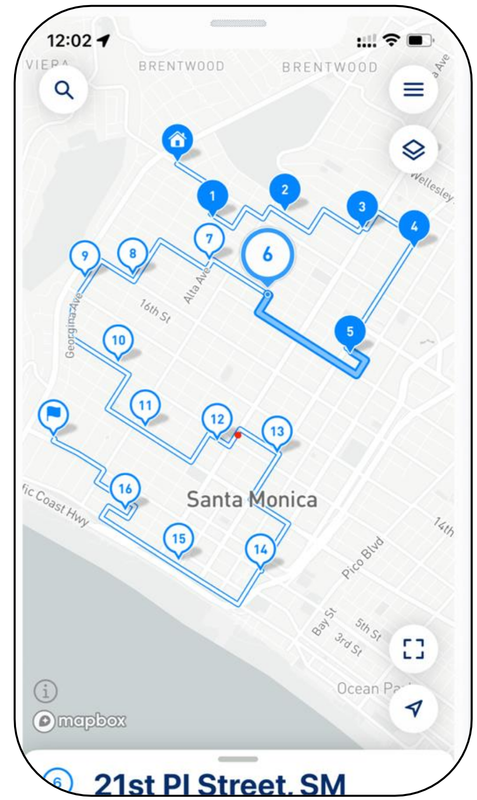
Ex) Uber, Lyft

Ridesharing route optimization



Ex) Uber, Lyft

Delivery route optimization



Ex) Amazon, Walmart

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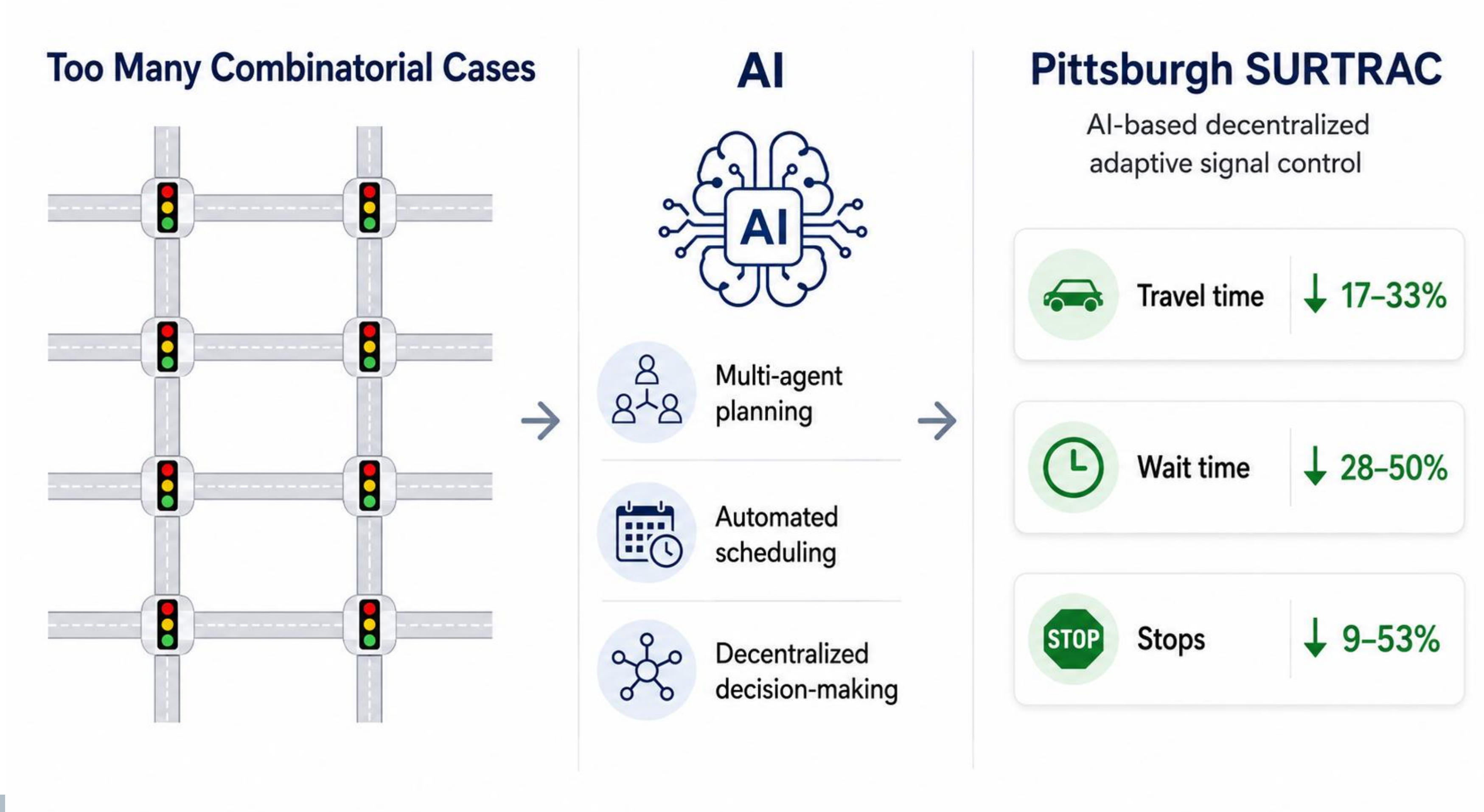
AI in Transportation Engineering

- **Example: AI-Enhanced Signal Control**



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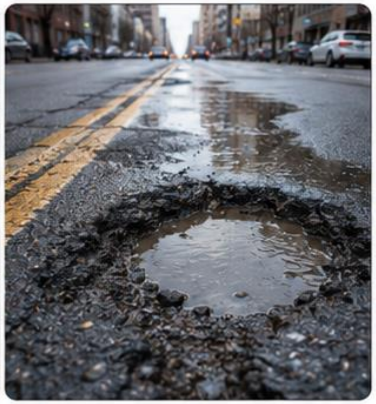
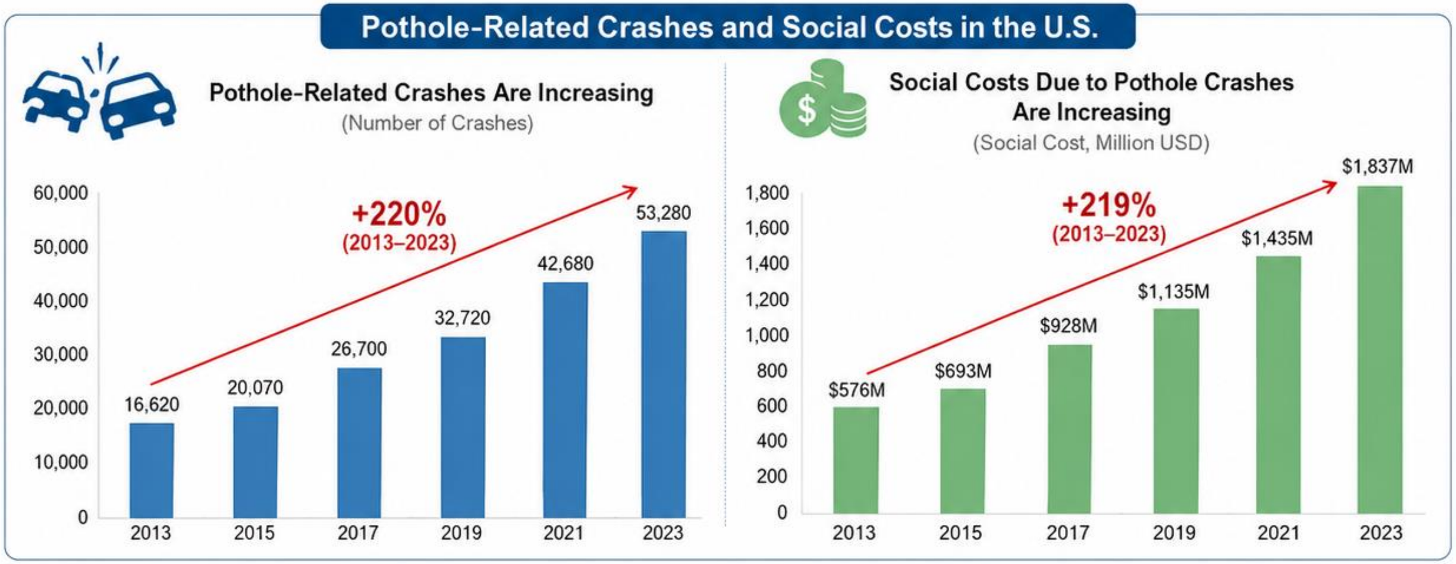
Infrastructure Monitoring and Predictive Maintenance

- ❑ **Road Pavement Monitoring**





Source: AAA Foundation for Traffic Safety (2023). "In the Driveway Potholes"; TRIP (2023). "The Economic Impact of Poor Road Conditions"

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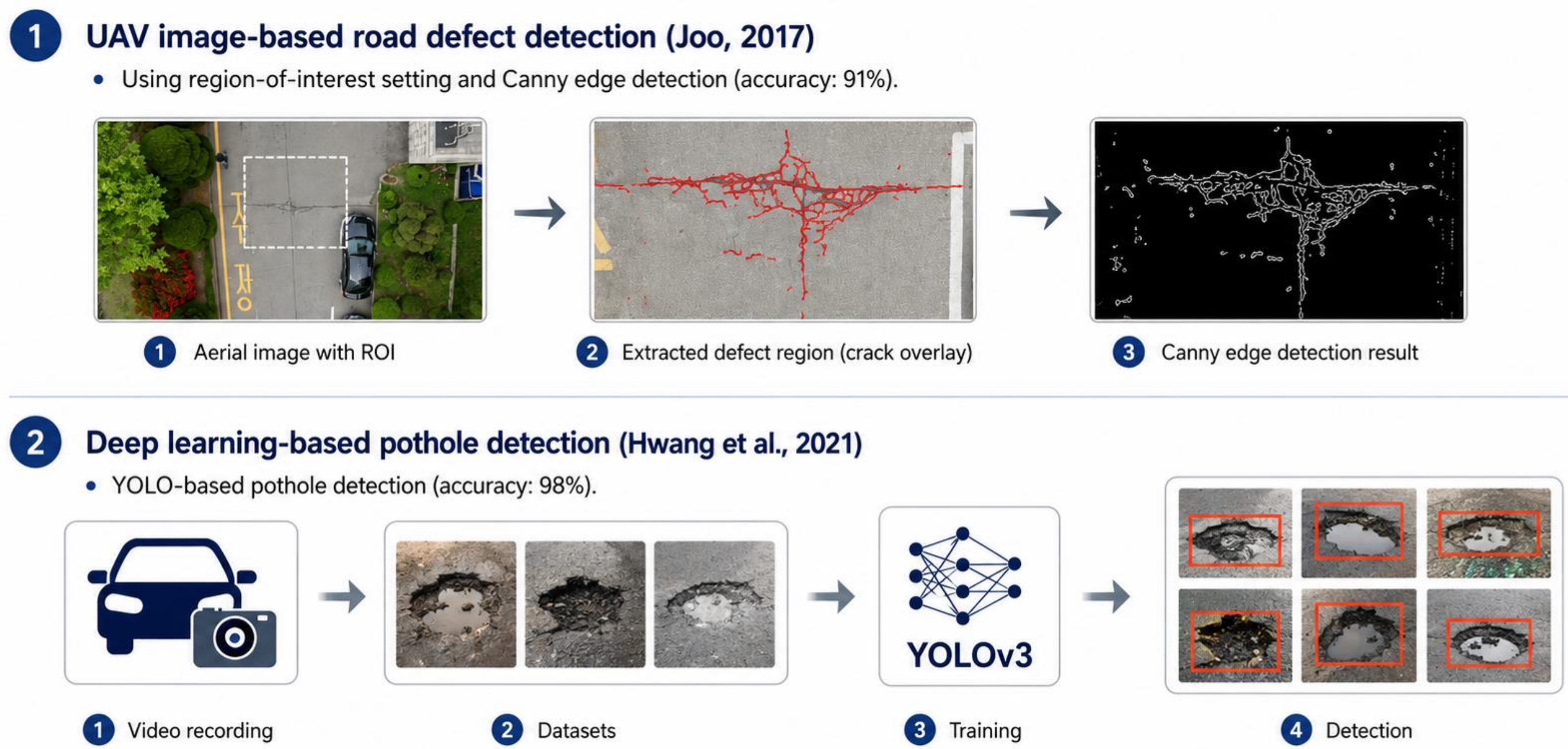
Infrastructure Monitoring and Predictive Maintenance

❑ Image-based Road Defect Detection Technologies



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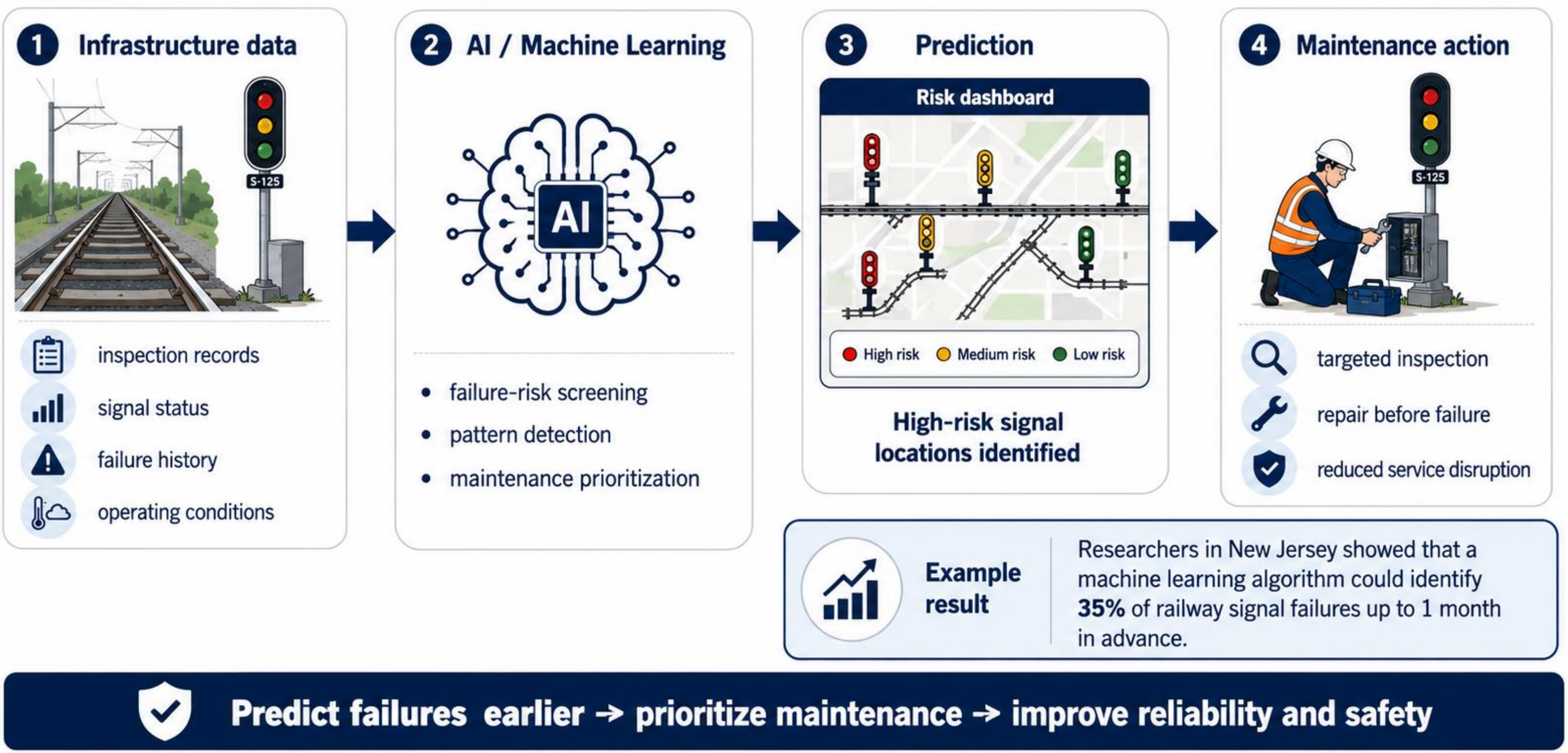
Infrastructure Monitoring and Predictive Maintenance

❑ **Railway Signal Predictive Maintenance**



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Construction Automation and Safety

- ❑ **Computer Vision for Construction**